# TraceAI: Dev Stack Execution Guide

A professional standard-operating procedure (SOP) for setting up, executing, and running the distributed Agentic & GenAI Platform locally.

## 1. Technical Overview

This document provides detailed setup pathways and multi-terminal orchestration instructions for compiling, running, and validating the TraceAI application. The system comprises a FastAPI backend engine, background Celery operations worker, local MongoDB, L1/L2 Redis caching instances, Ollama server instance, and Vite/React browser client.

| Component | Work Directory | Service/Port |
| --- | --- | --- |
| Docker Infrastructure | D:\GenAI and AgenticAI\genai-agent-sprint\backend | MongoDB (27017), Redis (6379) |
| Ollama Model Engine | Any (Host level / services) | Host daemon (11434) / llama3:8b |
| FastAPI API Backend | D:\GenAI and AgenticAI\genai-agent-sprint\backend | Uvicorn Daemon (8000) |
| Celery Work Engine | D:\GenAI and AgenticAI\genai-agent-sprint\backend | Celery Worker Thread (N/A) |
| Vite Client Frontend | D:\GenAI and AgenticAI\genai-agent-sprint\frontend | Vite Dev Server (5173) |

## 2. Prerequisites

- **Docker Desktop:** Running and configured for Linux container engines.
- **Python 3.12/3.13:** Installed with local paths mapped to environment variables.
- **Ollama:** Running locally serving port 11434 with llama3:8b-instruct-q4_K_M context downloaded.
- **NodeJS & NPM:** Stable LTS Node environment configured.

## 3. Multi-Terminal Orchestration (5 Terminals Required)

To execute the complete application stack, you must spawn and keep open **5 logical terminal sessions**. The step-by-step workflow is detailed below:

### Terminal 1: Spin Up Infrastructure Containers

```
Path: D:\GenAI and AgenticAI\genai-agent-sprint\backend
Command:
docker compose up -d mongo redis
```

### Terminal 2: Start LLM Engine Daemon

```
Path: Any directory (global command interface)
Command:
ollama serve
ollama pull llama3:8b-instruct-q4_K_M
```

### Terminal 3: Start FastAPI Application Server

```
Path: D:\GenAI and AgenticAI\genai-agent-sprint\backend
Command:
..\.venv\Scripts\Activate.ps1
uvicorn app.api_app:app --reload --host 0.0.0.0 --port 8000
```

### Terminal 4: Start Celery Worker Daemon

```
Path: D:\GenAI and AgenticAI\genai-agent-sprint\backend
Command:
..\.venv\Scripts\Activate.ps1
celery -A app.infra.celery_app worker --loglevel=info --concurrency=1
```

### Terminal 5: Start Vite Client SPA

```
Path: D:\GenAI and AgenticAI\genai-agent-sprint\frontend
Command:
npm install
npm run dev
```

## 4. Health Check Verification Matrix

Once all 5 terminals are active, verify the services using the following URLs:

| Service Endpoint | Verification URL | Expected Behavior |
|---|---|---|
| FastAPI Swagger Docs | http://localhost:8000/docs | Returns API list with green 200 OK responses. |
| Vite Web Portal Client | http://localhost:5173 | Opens TraceAI Agent Platform registration page. |
| Ollama Daemon Engine | http://localhost:11434 | Returns 'Ollama is running'. |